An Explicit Separation Between Head Complexity and Threshold Degree

A Walsh bilinear construction

Result. There is an explicit Boolean function f on 128 variables for which

$$\deg_{\pm}(f) = 2 \qquad \text{but} \qquad H^*(f) \geq 3.$$

Hence $H^*(f) \neq \deg_{\pm}(f)$. Here $\deg_{\pm}$ is threshold degree, also called sign degree.

1. The construction

Let $U = \{0,1\}^6$. Introduce two blocks of Boolean variables

$$x = (x_u)_{u \in U}, \qquad y = (y_v)_{v \in U}.$$

Each block has 64 variables. For $u, v \in U$, define the Walsh sign

$$W_{u,v} = (-1)^{\langle u,v \rangle_2}, \qquad \langle u, v \rangle_2 = \sum_{k=1}^{6} u_k v_k \pmod 2.$$

Equivalently, in lexicographic order, $W = H_2^{\otimes 6}$ where

$$H_2 = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

Now set

$$P(x, y) = \frac{1}{2} + \sum_{u,v \in U} W_{u,v} x_u y_v, \qquad f(x, y) = \mathbf{1}\{P(x, y) > 0\}.$$

This formula is completely explicit. It assigns a sign to every one of the 2^{128} Boolean inputs.

For a concrete evaluation, take $u = 101100$ and $v = 011010$. Their mod 2 inner product is 1, so $W_{u,v} = -1$. On the input with only $x_u = y_v = 1$, the value of P is $-1/2$, and therefore $f = 0$. If instead $u = 101100$ and $v = 010010$, the inner product is 0, so $P = 3/2$ and $f = 1$.

2. The slice that exposes the separation

For each $u, v \in U$, let $z^{u,v}$ be the input on which $x_u = y_v = 1$ and all other variables are zero. Then

$$P(z^{u,v}) = \frac{1}{2} + W_{u,v}.$$

Thus the sign matrix of f on this 64×64 singleton slice is exactly W:

$$\operatorname{sign}(P(z^{u,v})) = W_{u,v}.$$

The rest of the argument compares two ways of realizing this sign matrix.

3. Its threshold degree is exactly two

The bilinear polynomial P has degree 2. The sum in its definition is an integer on every Boolean input, so P is always a nonzero half-integer. Consequently P strictly sign-represents f, and

$$\deg_\pm(f) \le 2.$$

To rule out degree 1, recall that the sign rank $\mathrm{srank}(S)$ of a sign matrix S is the least rank of a real matrix having the same strict entrywise signs as S.

The Walsh matrix satisfies

$$WW^\mathsf{T} = 64I.$$

Indeed, distinct rows are orthogonal characters of $\{0,1\}^6$. Hence $\|W\|_2 = 8$. Forster's spectral norm bound gives

$$\mathrm{srank}(W) \ge \frac{\sqrt{64 \cdot 64}}{\|W\|_2} = \frac{64}{8} = 8.$$

Suppose that an affine function L sign-represented f. On the singleton slice it would have the form

$$L(z^{u,v}) = \lambda + \alpha_u + \beta_v.$$

The matrix $[L(z^{u,v})]_{u,v}$ has rank at most 2, since all its columns lie in the span of the all ones vector and $(\alpha_u)_u$. Its sign pattern would be W, contradicting $\mathrm{srank}(W) \ge 8$. Therefore no affine polynomial sign represents f, and

$$\boxed{\deg_\pm(f) = 2.}$$

4. Why two attention heads cannot compute it

For the one-layer attention model, the scalar contribution of one head is a ratio

$$\frac{N(x,y)}{D(x,y)},$$

where N and D are affine functions and D is positive on the Boolean cube. Thus a two-head classifier has a strict separating score of the form

$$S = c + \frac{N_1}{D_1} + \frac{N_2}{D_2}, \qquad D_1, D_2 > 0.$$

Clearing the positive denominators preserves every sign. The resulting polynomial is

$$Q = (N_1 + cD_1)D_2 + N_2 D_1.$$

In particular, Q is a sum of two products of affine functions.

We now bound the rank of any such polynomial on the singleton slice. Consider one affine product FG. On $z^{u,v}$, write

$$F(z^{u,v}) = \alpha + p_u + q_v, \qquad G(z^{u,v}) = \beta + r_u + s_v.$$

Its matrix of values decomposes as

$$F(z^{u,v})G(z^{u,v}) = R_u + C_v + p_u s_v + r_u q_v,$$

where

$$R_u = (\alpha + p_u)(\beta + r_u), \qquad C_v = \alpha s_v + \beta q_v + q_v s_v.$$

The term R_u is row-only and has matrix rank at most 1. The term C_v is column-only and also has rank at most 1. The final two terms are outer products, each of rank at most 1.

When two affine products are added, their two row-only terms combine into a single rank-one matrix, and their two column-only terms do the same. Four outer products remain. Therefore

$$\operatorname{rank}\big([Q(z^{u,v})]_{u,v}\big) \le 1 + 1 + 4 = 6.$$

Because Q has the same strict signs as the classifier, every function computable with two heads has singleton slice sign rank at most 6.

Our function has singleton slice sign matrix W, and $\operatorname{srank}(W) \ge 8$. This contradiction proves that two heads cannot compute it:

$$\boxed{H^*(f) \ge 3.}$$

5. Conclusion

For the explicit function

$$f(x,y) = \mathbf{1}\left\{\frac{1}{2} + \sum_{u,v \in \{0,1\}^6} (-1)^{\langle u,v\rangle_2} x_u y_v > 0\right\},$$

we have

$$\deg_\pm(f) = 2 < 3 \le H^*(f).$$

Thus threshold degree gives a universal lower bound on head complexity, but it need not equal head complexity.

The sign rank estimate uses J. Forster, “A linear lower bound on the unbounded error probabilistic communication complexity,” *Journal of Computer and System Sciences* 65 (2002), 612-625. doi:10.1016/S0022-0000(02)00019-3.